

DESCRIPTION

Tyzor® IAM is specifically developed for the printing ink industry. It is a titanium-based phosphate complex that offers the following features:

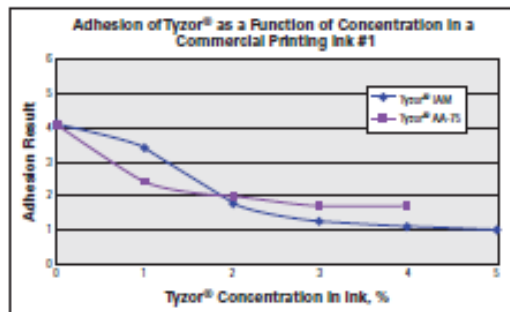
- Good stability in solvent-borne printing inks
- Good adhesion promotion and cross-linking properties resulting in improved water, solvent, and heat resistance
- High reactivity during curing
- Low or no discoloration
- No release of acetylacetone
- Effective at low concentrations (1–4%)

FUNCTIONALITY

Ink #1: Adhesion of printing ink to OPP foil as a function of titanate concentration

Procedure: Tyzor® IAM or Tyzor® AA-75 was added in different amounts to a commercial NC printing ink. The inks were applied and dried at room temperature. Adhesion was measured after 1, 3, 5, 10, 30 minutes, and after 1, 3, 5, 24 hours using adhesive tape pull off.

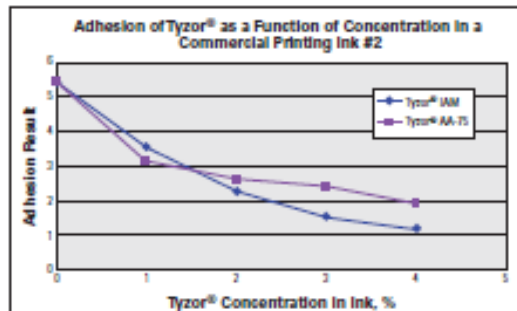
Results: The results are ranked on a scale of 1 to 6, with 1 representing very good adhesion (no delamination) and 6 corresponding to very poor adhesion (full delamination). The following charts compare the adhesion of the ink containing Tyzor® IAM vs. ink containing Tyzor® AA-75. In the first two charts, Tyzor® IAM shows better adhesion at higher concentration (> 2%) compared with Tyzor® AA-75. The third chart shows that the adhesion performance of Tyzor® IAM and Tyzor® AA-75 in a basic formulation were similar.



Ink #2: Solvent/Water Resistance

Procedure: Tyzor® IAM and Tyzor® AA-75 were added in different amounts to a commercial NC printing ink. The ink was applied and dried at room temperature. Solvent resistance was measured by applying various 50/50 and 70/30 ethanol/water solutions to the ink for 1 min.

Results: The results showed no significant difference between Tyzor® IAM and Tyzor® AA-75 in solvent or water resistance. Tyzor® IAM consistently produces very good results.



HOW TO USE

A Basic Formulation Contains:

Ethanol	32.4 pbw
Ethylacetate	10.0 pbw
Plasticizer	5.6 pbw
Pigment (TiO ₂)	40.0 pbw
NC Resin (35% a.i.)	12.0 pbw
Total	100 pbw

TYPICAL PROPERTIES

PROPERTY

TYPICAL VALUE

Form	liquid
Color	colorless to light yellow
TiO ₂ content	ca. 14.7%
Odor	sweet
Pour point	< -50°C
Boiling point	80°C
Flash point	12°C
Specific gravity	ca. 1.0 (25°C)
Viscosity	ca. 20 mPa•s (25°C)
Solubility in water	decomposes in water
Please refer to "Tyzor Global Registration Information" Bulletin	

GLOBAL REGISTRATION INFORMATION

SAFETY and HANDLING

For specific safety, handling and toxicity information, please refer to the current Material Safety Data Sheet.

TYPICAL SHELF LIFE

2 years

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